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Final Year Project – Report

Cascading electrochemical Brayton cycles for high-performance heat pumping

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Abstract

The UK projects a tenfold increase in heat pumps deployment by 2035 to address the challenges of decarbonising the heating sector. The dominant technology, vapour compression heat pumps, faces two structural limitations: the use of high global-warming-potential (GWP) refrigerants and a low efficiency ceiling due to operations on the reversed Rankine cycle. Brayton electrochemical heat pumps have emerged as a promising alternative, as they address both limitations simultaneously. However, a fundamental efficiency-power trade-off limits the proposed units' performance.

A Python model for ideal and non-ideal Brayton electrochemical single-cycle units is validated and used to characterise how the figure of merit ϕ , the current I , and the temperature difference across the cycle ΔT govern performance. The main sources of irreversibility identified from the model are deviations from isothermal heat exchange in ideal units, and overpotential losses in the non-ideal units. Multi-objective optimisation confirms the trade-off and identifies low ϕ , low I , and low ΔT as the desirable operating ranges. A cascaded-cycle unit is then designed to enable operations within these ranges and is modelled using the validated ideal single-cycle units. The performance of the novel design is compared to that of an equivalent single-cycle unit.

At a representative 12 K temperature lift, the cascaded unit achieves heating power increases of up to +365% with COP reductions of only 2% compared to the equivalent single-cycle unit. The cascaded unit also achieves ideal absolute COP around 18, benefiting from the high COP ceiling of the Carnot cycle. The cascaded architecture unlocks operations at lower ϕ and I ranges, inaccessible to equivalent single-cycle units. These results suggest that cascaded electrochemical Brayton cycles could deliver domestic heating at efficiencies more than three times those of current vapour compression heat pumps, directly addressing the requirement for low-GWP refrigerants and achieving the high COP needed to reduce peak electricity demand from mass heat pump adoption.

Nomenclature

a	Electrode Specific Surface Area
C_O	Oxidant Concentration
C_R	Reducing Species Concentration
C_p	Specific Heat Coefficient at constant pressure
C_v	Specific Heat Coefficient at constant volume
L_e	Length of the electrode
R_{mem}	Membrane equivalent resistance
T_C	Cold Reservoir Temperature
T_H	Hot Reservoir Temperature
$\dot{V}$	Volumetric flow rate
W_e	Width of electrode
e^-	Electron
F	Faraday's constant
I	Current
j	Current density
n	Electron Transferred per reaction
O	Oxidiser
q	Specific Heat
R	Reducing Species
R	Universal Gas Constant
T	Temperature
V	Potential
w	Specific work
ΔT	Temperature difference
S	Entropy

Greek letters

α	Temperature Coefficient
ε	Efficiency
η	Overpotential
ξ	Reaction coordinate
ϕ	Non dimensional performance variable

Abbreviations

ASHP	Air Source Heat Pump
BECR	Brayton Electrochemical Refrigeration
COP	Coefficient of Performance
GWP	Global Warming Potential
HCFC	Hydrochlorofluorocarbon
HFC	Hydrofluorocarbon
HVAC	Heating, Ventilation, and Air Conditioning

HX	Heat Exchanger
VCHP	Vapour Compression Heat Pump

Subscripts

act	Activation
actual	Related to the real process
BV	Butler Volmer
c	Cold, absorption side
Carnot	Relative to the Carnot cycle
CELL	Relative to the cathode + anode system (=cell)
cold	Related to the heat absorption process
conc	Concentration
E	Entropic forces
e	Electrolyte
h	Hot, rejection side
hot	Related to the heat rejection process
I	Thermal inertia
max	Maximum
N	Number of cycles in the cascaded configuration
ohmic	Ohmic
r	ratio of hot and cold
Ratio	Quotient of absolute to Carnot
rev	Reversible
rxn	Related to the total reaction
SET	Temperature wanted by the user (T_H) for cooling
useful	Rejected or absorbed based on heating or cooling configuration (respectively)

Superscripts

B	Bulk
S	Surface

States

1	Cold electrochemical cell inlet
2	Cold heat exchanger inlet
3	Hot electrochemical cell inlet
4	Hot heat exchanger inlet

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1. Introduction

1.1 Context and Motivation

The heating sector accounts for over 40% of domestic energy consumption in the UK and it mostly relies on natural gas boilers as the primary energy source (79%) [1], making it simultaneously the largest energy end-use and one of the hardest sectors to decarbonise. At present, the technology cannot reliably fit in a net-zero path without substituting natural gas as a fuel. Heat pumps offer the most viable alternative: by exploiting low-grade thermal energy from the environment, they deliver more than 2 units of heat per unit of electrical work input as determined by the Coefficient of Performance (COP) [2], and studies show that replacing gas boilers with heat pumps is one of the most promising paths to decarbonise the heating sector [3]. Recognising this, the UK projects a cumulative domestic heat pump deployment of 12 million units in 2035, at a minimum rate of 600,000 per year by 2028, a 10-fold increase in ten years (from the 1.3 million in 2025) [3], [4]. Meeting this deployment target, while simultaneously reducing whole-life carbon impact, demands heat pump systems free from the high global-warming-potential (GWP) refrigerants currently in use and more efficient designs to avoid disruptive increases in peak electricity demand [5].

In fact, current vapour-compression heat pumps (VCHP) present two structural problems that incremental engineering improvements cannot resolve. First, the hydrofluorocarbon based refrigerants on which the technology depends have GWP orders of magnitude greater than CO₂ (R134a's GWP for example is 1530 times CO₂'s GWP [2]) and, while the Montreal Protocol entered into force in 2019 to phase these down [6], no viable drop-in replacement at scale has yet been identified. Second, and more fundamentally, vapour-compression systems operate on the reversed Rankine cycle, which carries an inherently lower maximum efficiency ceiling than the reversed Carnot cycle [7] causing the absolute COP to be relatively low.

Electrochemical heat pump (ECHP) cycles address both limitations simultaneously and for this reason they stand-out among the alternative technologies to VCHPs. They exploit the entropy changes associated with reversible electrochemical reactions to absorb and reject heat [8]. These systems require no high-GWP refrigerants and, operating on thermodynamic cycles that can theoretically approach Carnot efficiency, face no equivalent ceiling to that of the reversed Rankine cycle [7]. These properties position ECHPs as a structurally superior alternative to VCHP and make them an important research area.

The following section covers the state of research on electrochemical systems in heating and identifies the research gaps that this work aims to address through the modelling of a novel electrochemical unit using cascaded cycles to enhance unit's performance.

1.2 Literature Review

"Not-in-kind" heat pump technologies have emerged in recent years as a key alternative to VCHPs [9]. Zhu et al. [10] provide a comprehensive classification of such systems and, among these, electrochemical heat pumps are the focus of this study. The electrochemical systems are further split by Zhu et al. [10] into indirect and direct configurations. Indirect electrochemical systems couple electrochemical processes with secondary thermodynamic cycles using interconvertible working fluid pairs such as *isopropanol* $\leftrightarrow$ *acetone* [11], and include metal hydride heat pumps [12] and electrochemical looping systems. Direct systems share a common working principle: entropy changes associated with reversible electrochemical reactions absorb and reject heat at the electrode surfaces, with the direction of heat transfer controlled by the sign of the applied voltage [8]. The key parameter governing this effect is the temperature coefficient $\alpha = \partial V / \partial T = \Delta S_{rxn} / nF$, which quantifies the sensitivity of electrode potential to temperature and directly determines the magnitude of the heating or cooling effect achievable [13]. Zhu et al. [10] report that direct electrochemical systems can achieve COPs of 3.5-14.3 under standard operating conditions: a large efficiency advantage over conventional VCHPs. Three direct cycle configurations are reviewed below.

Stirling electrochemical refrigeration cycles

The Stirling electrochemical refrigeration cycle (example in Figure 1) consists of two cells (endothermic and exothermic) each paired with a heat exchanger and connected via an electrolyte flow circuit which includes a regenerator, reservoirs, and pumps [7], [14], [15].

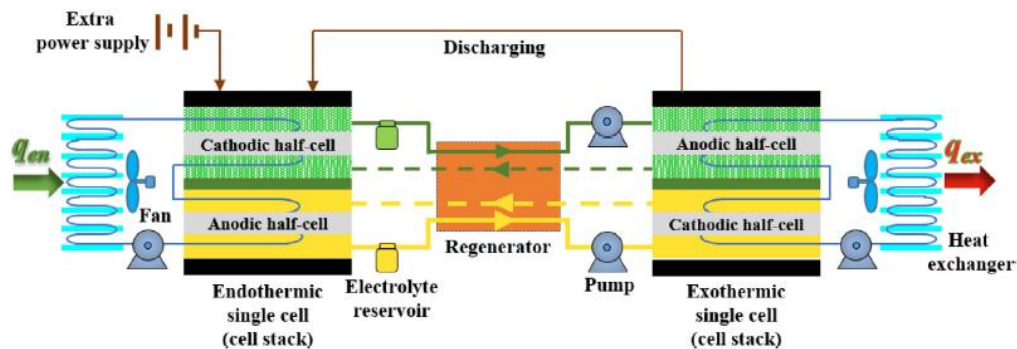


Figure 1 Stirling cycle-based system, schematic from [7]

Heat absorption occurs through redox reactions in the endothermic cell, while the opposing reactions in the exothermic cell cause heat release. Duan et al. [7] presented the highest performing Stirling-based model: a continuous iron-vanadium system with COP values of 2 to 6.5 across current densities of 10 to 200 A/m^2 . However, in this type of systems, the isothermal heat absorption process is complicated by the advecting electrolyte simultaneously exchanging heat with the surroundings and resulting in a coupled heat and mass transfer problem that limits practical performance and directly motivates the development of alternative technologies that avoid this type of operations: the BECR cycle [13].

Brayton ElectroChemical Refrigeration (BECR) cycles

The BECR cycle is the highest-performing direct electrochemical system in the literature. Like the Stirling cycle, it exploits entropy changes in reversible redox reactions but adopts a cycle architecture in which the electrochemical cells are thermally isolated from the heat exchangers [13], [16], separating the reaction and heat transfer processes that the Stirling cycle couples together limiting its practical performance. A schematic is shown in Figure 2.

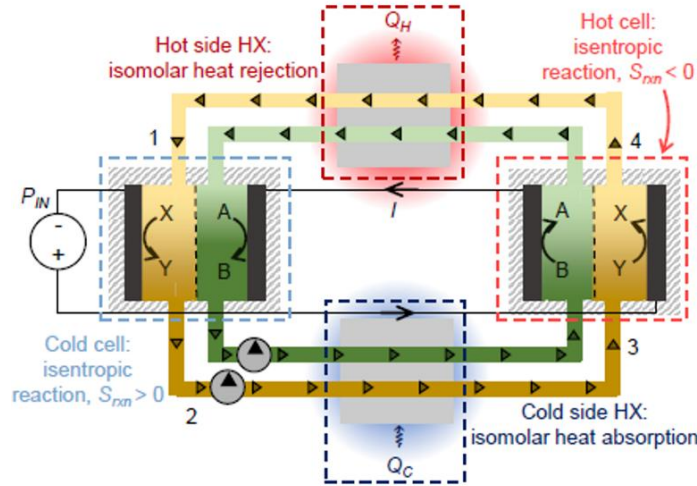


Figure 2 BECR unit schematic from [16]

The system comprises two pairs of adiabatic half-cells (with opposing temperature coefficients α) through which electrolytes circulate in a four-process closed cycle: isentropic cooling via endothermic reaction, heat absorption from the cold reservoir, isentropic heating via exothermic reaction, and heat rejection to the hot reservoir. The continuous operation of this cycle is made possible by employing all-soluble redox couples and flow battery cell architectures, which allow the electrolytes to be pumped between components. Depending on the definition of the surrounding environments as cold or hot reservoirs, the cycle can operate as either a refrigerator or a heat pump.

Rajan et al. [13] provided the first experimental proof of concept of a continuously operating BECR cycle, using the I^-/I_3^- and FCN^{3-}/FCN^{4-} redox couples, chosen for their relatively high combined cell temperature coefficient $\alpha \approx 2 \text{ mV/K}$. A peak COP of 8.09 was measured at a temperature difference between reservoirs of only 0.07 K, and a peak cooling load of 0.934 W was achieved at a COP of 0.93 with a temperature difference of 0.15 K. The COP was observed to drop steeply with increasing operating current as a consequence of Joule heating, which scales with I^2 and increasingly offsets the entropic cooling effect.

Subsequent electrolyte engineering using a synergistic perchlorate anion and acetonitrile co-solvent strategy (which selectively rearranges the Fe^{2+} solvation shell to maximise the entropy difference between oxidation states) managed to raise α to 3.73 mV/K while

simultaneously reducing specific heat capacity, yielding a demonstrated electrolyte temperature drop of 1.42 K and a potential COP of 14.3 [17].

Rajan et al. [16] developed a comprehensive framework for the BECR cycle, introducing a dimensionless figure of merit χ (called ϕ in this work), representing the ratio of the entropic driving force to the thermal inertia of the electrolyte. The study of ideal systems showed that COP decreases monotonically with increasing χ , while cooling load first increases and then decreases, with peak χ that can be identified analytically. The analysis also demonstrated that regeneration (defined as internal heat exchange between the electrolyte at the inlets of the cells) relaxes the temperature constraints imposed by the source and sink and simultaneously increases both COP and cooling load for a given χ . For the non-ideal cycle, the study included activation, concentration, and ohmic overpotentials derived from Butler-Volmer equation in a one-dimensional adiabatic plug flow cell model as a function of current. Ohmic overpotential is identified as the dominant source of heating in both the hot and cold cells.

It is important to notice that the adiabatic plug flow cell model code provided as supplementary material in [16] presents minor inconsistencies and has to be corrected.

The low temperature lifts achievable for the presented units directly motivate the development and assessment of new systems to resolve this limitation. The development of cascaded cycle is grounded in this motivation.

Cascaded BECR

As reviewed above, for single BECR cycles, high COP values are achieved only at small temperature lifts, well below the 20 to 30 K required for domestic heating applications. Sha et al. [18] identified that cascading multiple BECR units in series addresses this issue by exploiting the cumulative temperature lift across stages. Sha et al. [18] proposed and modelled a cascade electrochemical Brayton refrigeration cycle and demonstrated that, with six cascaded cycles, temperature lifts can reach up to 109.8 K compared to 16.3 K for single cycle equivalents. However, the gains follow a law of diminishing returns: the increase in temperature lift between successive cells drops from 15.2% (2 to 3 cells) to 1.3% (5 to 6 cells), while net electrical work input grows disproportionately, causing COP to fall from 8.44 with 2 cells to 2.73 with 6 cells [18]. The same study further identifies α as the most influential parameter, with increases yielding up to 328% improvement in temperature lift, while heat recovery efficiency is critical: at zero recovery, cascading provides no temperature gain regardless of cell number. These findings establish the starting point of this project, inspiring the use of cascaded designs not only to address temperature lifts, but to enhance the power of units while retaining the COP performance, solving also the second limitation of the single-cycle units.

Despite the promising technology, existing electrochemical cycle studies remain far from domestic applicability, either presenting low temperature lifts at high COP values [10] or very low power at acceptable COP and temperature lifts [19]. While high COP values are desirable, they are only meaningful if the unit delivers sufficient power to maintain the set temperature, and a unit operating at near-zero power output has no practical value regardless of its COP. While cascading multiple BECR units can enhance temperature lift significantly [18], no study has assessed the effect of temperature lift across a cell, mapped the COP-heating power trade-off in single-cycle units across relevant operating ranges, or proposed a cascaded design to enhance power and mitigate the trade-off effects.

1.3 Aim and Objectives

This project addresses the research gap identified above. The aim is to develop a Python model of a cascaded-cycles unit to enhance the unit's power and address its performance across the desired conditions compared to the equivalent single-cycle unit. The design should be independent of the use of high-GWP refrigerants and with better energy performance compared to the industry standard to address the pressing need for alternative solutions identified in the first section. To achieve this goal, the research questions that this project tries to answer are the following:

1. What are the performance limitations of the non-ideal single-cycle unit and the consequent desired ranges of operating parameters?
2. Can cascaded-cycles units allow operation in the desired operating ranges to achieve enhanced performance?
3. How does the new unit design compare to the equivalent single-cycle unit?

The steps to answer these questions include modelling and validating the ideal and non-ideal single-cycle units and using them to identify the factors that limit performance, using the knowledge from the single-cycle unit investigation to design a cascaded-cycle unit that operates in the desired ranges, and lastly evaluating the performance (quantified as COP and heating power) of the equivalent single-cycle unit and comparing it to the performance of the novel design.

2. Methodology

2.1 Physical Processes

Both the single-cycle and the cascaded-cycle designs are based on the electrochemical unit (or a cascade of it) shown in Figure 3 and operating on the Brayton cycle [16]. The unit includes two adiabatic electrochemical cells, two heat exchangers (HX), electrolyte pumps, and power sources. The processes are undertaken by a general electrochemical system including two redox-active species, the electrodes, and the ion-exchange membrane. The streams are referred to as “the electrolyte” with average properties and states of the flows.

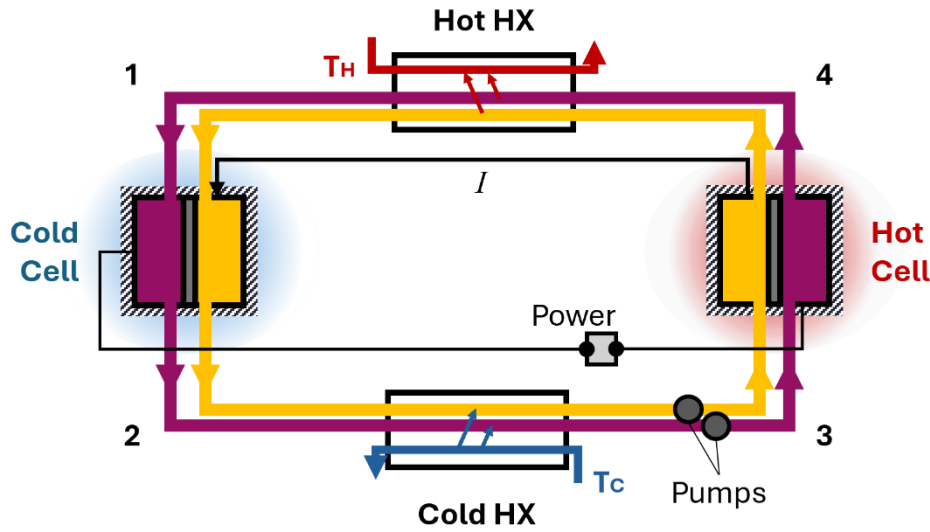


Figure 3 Single-cycle unit schematic

Thermogalvanic effects in the cells allow temperature variations of the electrolyte (as will be described in later sections) and are quantified via the temperature coefficient $\alpha = \Delta S_{rxn}/nF$ for a general half-cell reaction $O + n e^- \rightarrow R$. Each of the unit's cells comprises two half-cell reactions with opposite signs of α , resulting in a maximised $\alpha_{CELL} = \alpha_{cathode} - \alpha_{anode}$. The value of α_{CELL} directly determines ΔS_{rxn} for a fixed reaction and, it will be shown, that this directly influences the temperature difference between the inlet and the outlet of the cell.

The electrolyte is also the working fluid and is circulated by two pumps (one per stream) through the four components to undergo the processes described here for an ideal unit:

- 1 to 2: Endothermic reactions in the cold cell require heat which can only be sourced from the electrolyte since the cell is adiabatic, causing the temperature of the electrolyte to decrease and end below the cold reservoir temperature (T_C);
- 2 to 3: Heat absorption from the environment in the cold HX heats the electrolyte to the cold reservoir temperature T_C (in the ideal case; Later considerations will address ε_{HX});
- 3 to 4: Exothermic reactions in the hot cell heat the electrolyte above the hot reservoir temperature (T_H) through a process analogous but opposite to that of the cold cell.

- 4 to 1: Heat rejection into the hot reservoir through the hot HX closes the cycle by cooling the electrolyte to T_H and provides the heating power demanded.

This specific configuration (presented by [16]) is chosen as the starting point of the project since it has demonstrated promising performance and qualities compared to other solutions.

To scale out power, the cascaded-cycle architecture (3 cycles example shown in Figure 4b) is proposed. In this unit, the hot HX of mid cycles is the cold HX of the warmer adjacent cycle, exchanging heat between electrolytes of two cycles rather than with an external reservoir. The cycles are all modelled to work at the same current. The equivalent single-cycle unit operates across the same ΔT and can either be one unit with the same maximum and minimum electrolyte temperatures as the cascaded cycle (equivalent $\phi = N \times \phi_{cascaded}$), or the unit with N single-cycles at the same ϕ and I as the cascaded unit (Figure 4a). The main difference between the cascaded and the single cycle solutions is the ΔT across each cycle: in the single cycle units it is $T_H - T_C$, while in the cascaded cycle it is only a fraction of $T_H - T_C$.

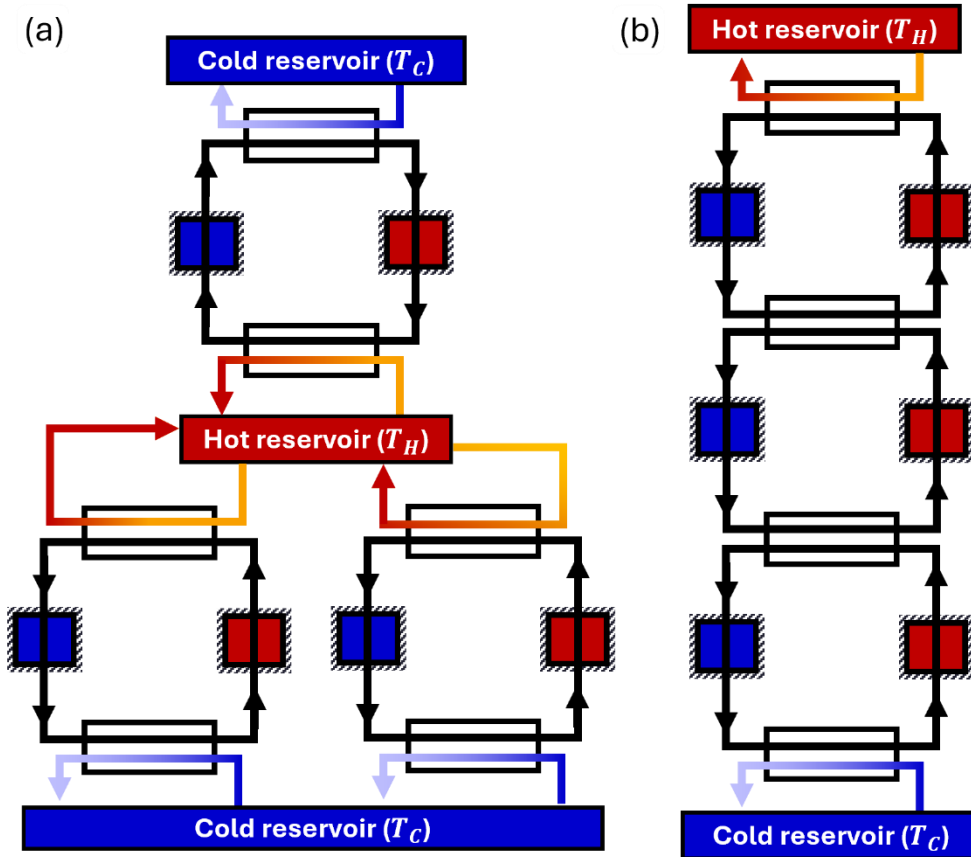


Figure 4 Cascaded-cycles configuration (b) vs. equivalent single-cycles (a)

2.2 Model equations

The equation governing the temperature behaviour of the electrolyte in the electrochemical cells is the total derivative of the entropy of an electrochemical system undergoing an isentropic reaction, Equation 2.1:

$$dS = C_p \frac{dT}{T} + \Delta S_{rxn} d\xi \quad (2.1)$$

where ξ is the reaction coordinate, S is entropy, ΔS_{rxn} is the reversible entropy change, T is temperature, and C_p is the specific heat at constant pressure. In the ideal electrochemical cells, the processes are isentropic. Therefore, the electrolyte's temperature variations (dT) are opposite to ΔS_{rxn} in sign. This means that for the endothermic ($\Delta S_{rxn} > 0$) reactions in the cold cell $dT < 0$, and for the exothermic ($\Delta S_{rxn} < 0$) reactions in the hot cell $dT > 0$.

The model used for the temperature evolution in the electrochemical cells is the adiabatic plug flow model [16]. A low order 1D model for half of the adiabatic plug flow electrochemical cell was developed by Rajan et al. [16] based on the local energy balance in an elemental volume of the domain. The domain of the model includes the current collector, a porous electrode and one half of the membrane.

Equation 2.2 is obtained:

$$\dot{V} C_v \frac{dT}{dx} = -\frac{I}{L_e} (\eta + T\alpha) \quad (2.2)$$

Where $\dot{V}$ is the volumetric flow rate of the electrolyte, C_v is the specific heat at constant volume, x is the direction of flow of the electrolyte and the discretisation direction of the domain, I is current, L_e is the length of the electrode in the direction of flow of the electrolyte, η is the signed overpotential, and α is the temperature coefficient as identified in Section 2.1.

2.2.1 Metrics and Figure of Merit

Before modelling the cycles under various assumptions, it is useful to define the performance metrics and the figures of merit used in the cycles. The two performance metrics assessed are COP and heating/cooling power, defined by Eqs. 2.3 and 2.4, respectively.

$$COP = \frac{q_{useful}}{w_{input}} = \begin{cases} \frac{q_{hot}}{q_{hot} - q_{cold}} & \text{if heating} \\ \frac{q_{cold}}{q_{hot} - q_{cold}} & \text{if cooling} \end{cases} \quad (2.3)$$

$$\dot{Q}_{useful} = \begin{cases} \dot{Q}_h = \dot{V}_e \times C_{v,e} \times (T_4 - T_1) & \text{if heating} \\ \dot{Q}_c = \dot{V}_e \times C_{v,e} \times (T_3 - T_2) & \text{if cooling} \end{cases} \quad (2.4)$$

The relevance of the absolute value of the COP is to be questioned. It is useful to quantify the absolute value in order to directly assess the efficiency of a technology. However, to compare technologies performances, it is better to normalise the COP by the theoretical maximum the cycle can achieve, in this case the Carnot cycle COP, defining Eq. 2.5:

$$COP_{ratio} = COP/COP_{Carnot} \quad (2.5)$$

where

$$COP_{Carnot} = \begin{cases} \frac{T_H}{T_H - T_C} & \text{if heating} \\ \frac{T_C}{T_H - T_C} & \text{if cooling} \end{cases} \quad (2.6)$$

Where T_H and T_C are the hot and cold reservoirs temperatures (respectively). To characterise operating conditions, it is useful to define temperature lift and heating temperature ratio:

$$\Delta T = T_H - T_C \quad (2.7)$$

$$T_r = T_H/T_C. \quad (2.8)$$

Lastly, an important figure of merit, ϕ , representing the ratio of entropic driving force (ϕ_E) to thermal inertia of the electrolyte (ϕ_I) emerges from the integration of Equation 2.2 under ideal operations:

$$\phi = \frac{I\alpha}{\dot{V}C_v} = \frac{\phi_E}{\phi_I} \quad (2.9)$$

2.2.2 Ideal single-cycle model

The ideal single-cycle model assumes reversible operation of both cells with no overpotential, therefore, Equation 2.2 can be further simplified by setting the overpotential losses $\eta = 0$. The equation can be rearranged and integrated:

$$\frac{\dot{V}C_v}{\bar{I}\alpha} \int_{T_{in}}^{T_{out}} \frac{1}{T} dT = -\frac{1}{L_e} \int_0^{L_e} dx \quad (2.10)$$

Assuming that the heat exchangers are ideal, which is equivalent to having an infinite heat exchange surface area so that the outlet temperature of a stream is equal to the inlet temperature of the other stream (with counterflow configuration), allows to set the completion of the heat exchange processes when the electrolyte reaches the temperature of the reservoir of reference ($T_1 = T_H$ and $T_3 = T_C$). Then, as the results of Equation 2.10, the temperature at state 2 and 4 can be written as:

$$T_2 = T_H \times \exp[-\phi] \text{ and } T_4 = T_C \times \exp[\phi]. \quad (2.11)$$

With these definitions of T_2 and T_4 , it is possible to express the performance parameters in terms of operating parameters and the figure of merit:

$$\dot{Q}_c = \frac{I\alpha T_c}{\phi} (1 - T_r \times e^{-\phi}) \quad (2.12)$$

$$\dot{Q}_h = \frac{I\alpha T_c}{\phi} (e^\phi - T_r) \quad (2.13)$$

$$COP_{heating} = \frac{e^\phi - T_r}{e^\phi + T_r \times e^{-\phi} - 1 - T_r} \quad (2.14)$$

2.2.3 Non-ideal single-cycle model

The real (or “non-ideal”) model introduces two kinds of losses: overpotentials and real heat exchanger efficiency. The total overpotential η is the sum of the contribution of the activation, concentration, and ohmic overpotentials (η_{act} , η_{conc} , η_{ohmic}). The first two are derived from the Butler-Volmer equation:

$$I = aV_e j_0 \left[\frac{C_O^S}{C_O^B} \exp\left(-\frac{\alpha_{BV} nF\eta}{RT}\right) - \frac{C_R^S}{C_R^B} \exp\left(\frac{(1 - \alpha_{BV}) nF\eta}{RT}\right) \right]. \quad (2.15)$$

Where a is the electrode specific surface area, α_{BV} is the symmetry coefficient, C is the concentration, F the Faraday’s constant, R the universal gas constant, and n the number of e^- exchanged in the reaction. The subscripts O and R relate to oxidising and reducing species, while the superscripts S and B refer to surface or bulk. The activation overpotential is derived by intentionally imposing uniform concentration thereby eliminating concentration overpotential, and by setting the coefficient of symmetry $\alpha_{BV} = 0.5$. Yielding:

$$I = aV_e j_0 \left[\exp\left(-\frac{nF\eta_{act}}{2RT}\right) - \exp\left(\frac{nF\eta_{act}}{2RT}\right) \right] \quad (2.16)$$

and recognising that $e^\theta - e^{-\theta} = 2\sinh(\theta)$, the equation becomes

$$I = aV_e j_0 \left[2\sinh\left(-\frac{nF\eta_{act}}{2RT}\right) \right]. \quad (2.17)$$

With $j_0 = nFk\sqrt{C_O^B C_R^B}$ and defining $j = \frac{I}{aV_e}$, the equation can be rearranged to isolate η_{act} as

$$\eta_{act} = -\frac{2RT}{nF} \sinh^{-1}\left(\frac{j}{2j_0}\right). \quad (2.18)$$

A similar approach is followed to determine the concentration overpotential. Starting with the intuition that η changes sign in the reduction and oxidation reactions (i.e. $\eta < 0$ in cathodes and $\eta > 0$ in anodes), it is clear that at high cathodic current densities $\exp(-nF\eta/2RT) \gg \exp(nF\eta/2RT)$ since $\exp(-nF(-|\eta|)/2RT)$ is increasing with η , while $\exp(nF(-|\eta|)/2RT)$ decreases with η . Vice versa, at high anodic current densities, $\exp(nF\eta/2RT) \gg \exp(-nF\eta/2RT)$. In these regimes, called the Tafel regimes, the concentration overpotential η_{conc} dominates and Eq. 2.15 can be simplified as

$$j = j_0 \left(\frac{C_O^S}{C_O^B} \right) \exp\left(-\frac{nF\eta_{conc}}{2RT}\right) \quad (2.19)$$

and the concentration overpotential is determined by

$$\eta_{conc} = \begin{cases} -\frac{2RT}{nF} \left[\ln \left(\frac{C_O^S}{C_O^B} \right) \right] & \text{at the cathode} \\ \frac{2RT}{nF} \left[\ln \left(\frac{C_R^S}{C_R^B} \right) \right] & \text{at the anode} \end{cases} \quad (2.20)$$

after incorporating the constant term $\ln(j_0/j)$ into the reference potential. This shows how concentration overpotentials become meaningful when the surface concentrations of a species deviate from its bulk concentration.

The ohmic effects in the electrolyte have been ignored as expected to be negligible compared to the membrane losses [16] with equivalent resistance R_{mem} . By assuming each electrode is responsible for half of the ohmic loss, the ohmic overpotential is defined as

$$\eta_{ohmic} = \frac{IR_{mem}}{2}. \quad (2.21)$$

Heat exchangers effectiveness is defined based on the temperature difference in each stream of fluid exchanging heat with the cold or hot reservoirs:

$$\varepsilon_{cold\ HX} = \frac{q_{c,actual}}{q_{max}} = \frac{T_3 - T_2}{T_C - T_2}, \quad (2.22)$$

$$\varepsilon_{hot\ HX} = \frac{q_{h,actual}}{q_{max}} = \frac{T_1 - T_4}{T_H - T_4}. \quad (2.23)$$

2.2.4 Cascaded-cycle model

The ideal cascaded-cycle model exploits common heat exchangers between consecutive cycles. The cycles are numbered from 1 to N starting from the cycle that absorbs heat from the cold reservoir, and a temperature notation is introduced: T_{nk} where n indicates the cycle number (from 1 to N) and k the state number (ranging from 1 to 4 as defined in Figure 3).

For cycle N , the hot heat exchanger is the same as a single-cycle and its effectiveness is described by Equation 2.23. Similarly, the cold heat exchanger of cycle 1 is the same as in a single-cycle and Equation 2.22 describes the effectiveness. In the ideal scenario, which was the only one considered in this work, both effectivenesses are set to 100%, leading to $T_{13} = T_C$ and $T_{N1} = T_H$ since the two external cycles exchange heat directly with the cold and hot reservoirs. For intermediate cycles, however, the effectiveness couples the temperatures of adjacent cycles according to Equations 2.24 and 2.25

$$\varepsilon_{coldHX\ n} = \frac{T_{n3} - T_{n2}}{T_{(n+1)4} - T_{n2}}, \quad (2.24)$$

$$\varepsilon_{hotHX\ n} = \frac{T_{n1} - T_{n4}}{T_{(n-1)2} - T_{n4}}. \quad (2.25)$$

and setting both to 100% leads to $T_{n3} = T_{(n+1)4}$ and $T_{n1} = T_{(n-1)2}$ with $n = [1, N]$.

2.3 Numerical Procedures and Model parameters

All models are coded in Python. The cascaded model code is provided in the Appendix.

2.3.1 Single cycle

To model the performance of the ideal single cycle unit the Equations 2.12 to 2.14 can be directly applied by fixing the operating parameters I , α , ϕ , and T_H , and by knowing the cold reservoir temperature (T_C). To analyse effects of ϕ and I , the performance can be assessed by sweeping across different values of the variables (taking into account that $\phi = \phi(I)$).

For the non-ideal cycle, the cells are discretised along the flow direction and the overpotentials from Equations 2.18, 2.20, 2.21 are evaluated throughout the cell's length. The temperatures at the outlet of the heat exchangers are determined using the effectiveness in Equations 2.22 and 2.23, with the iterative procedure in Figure 5 with convergence when the difference between T_1 and the guessed T_1 at the cycle beginning is less than 0.01 K.

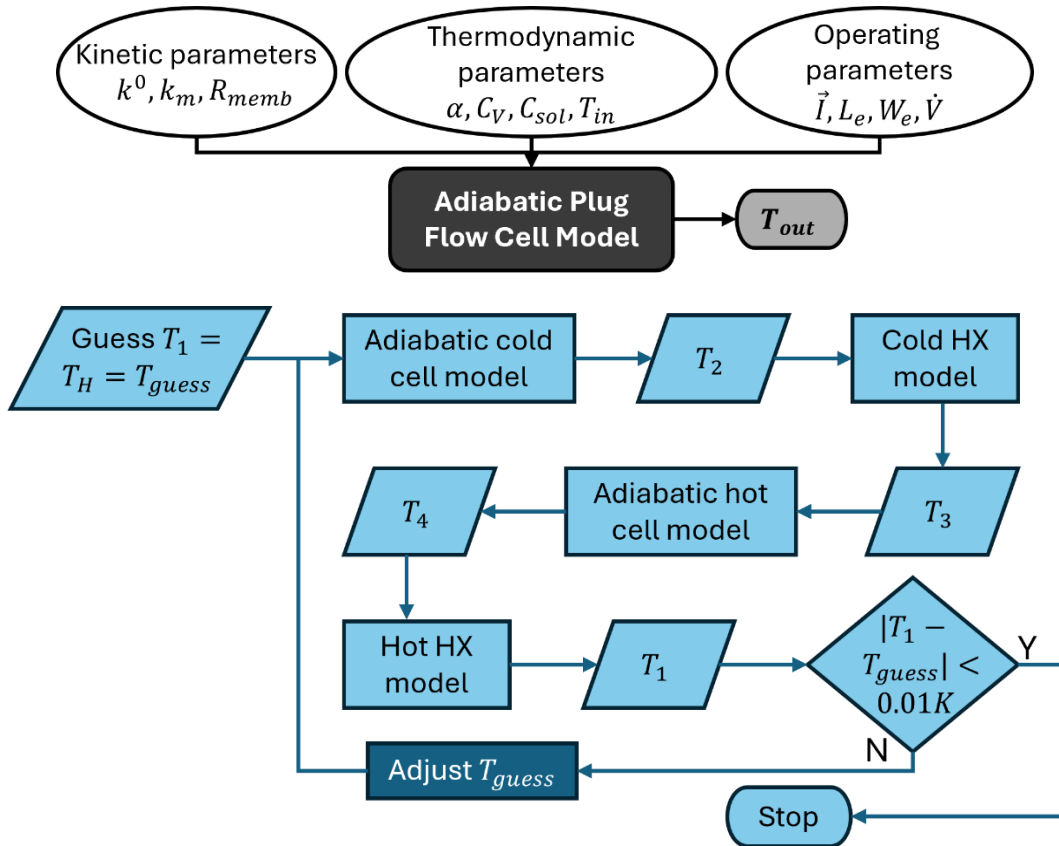


Figure 5 Iterative procedure to solve non-ideal cycle

Once the temperatures are converged, Equations 2.3 and 2.4 are used to evaluate the performance of the unit and sweep through a range of operating current values to assess the

effects of I on the performance metrics COP and Q_h . The performance is also analysed as a function of ΔT , by testing different values of T_C .

2.3.2 Cascaded cycle

The cascaded cycle configuration employs a two macro-steps solution process: the forward loop to solve for temperatures at states 1 and 2 of each cycle (Figure 6, purple blocks), and the backward loop to solve for temperatures at states 3 and 4 of each cycle (Figure 6, orange blocks). This solution procedure enables solving for all the coupled cycles with two sweeps and uses Equations 2.11 to 2.14 for the states and performance of each cycle by treating it as an ideal cell (according to Section 2.3.1).

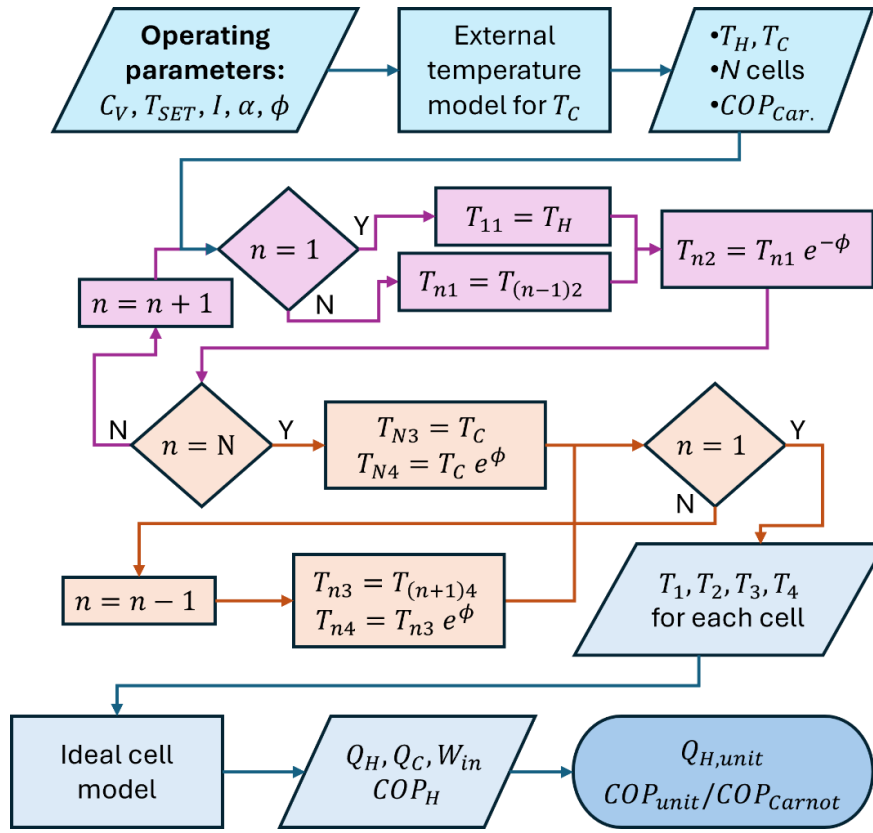


Figure 6 Solving procedure for the cascaded-cycle design

2.3.3 Input Parameters for the simulations

The input parameters for the simulations are reported in Table 1.

Table 1 Model parameters used in simulations

Symbol	Value	Unit	Reference
<i>Single-cycle simulations</i>			
T_H	313	K	[16]
T_C	295	K	[16]
α	1.5 for ideal model 2 for non-ideal model	mV/K	[16]
n	$1 e^-$		[16]

$C_{v,e}$	4.18 for ideal model 3.56 for non-ideal model	MJ/m^3K	[16]
ϕ	0.12		[16]
<i>Single-cycle simulations to investigate effects of ΔT</i>			
T_H	295 273	K	This work
ΔT	2, 4, 6, 8	K	This work
<i>Cascaded-cycle simulations</i>			
ϕ	0.0185		This work
I	1	A	This work
α	1	mV/K	This work
C_v	3.56	MJ/m^3K	[16]
T_H	293	K	This work

For the simulations in this work, the minimum T_C is selected at $0^\circ C$ considering the electrolyte to be liquid thanks to freezing point depression of aqueous electrolytes [20]. A maximum ΔT of $20^\circ C$ is set considering no heating demanded for temperatures $\geq 20^\circ C$.

2.3.4 Multi-objective optimisation

To map the simultaneous effects of multiple parameters on the COP-power trade-off, a multi-objective optimisation is performed on the non-ideal single-cycle unit using the NSGA-II algorithm implemented via the pymoo Python library. The variables swept are ΔT (from 6 to 20 K), I (0.01 to 10 A), α (1 to 5 mV/K), ε_{HX} (50 to 100%), and ϕ (0.1 to 1).

2.4 Validation Methods

The cascaded unit model is coded based on the single-cycle model, so it is crucial to validate the latter for reliable results. The validation is based on a comparison against literature results presented by Rajan et al. [16] for the same operations. The non-ideal model also needs validation against Rajan et al. [16] since the codes for the ideal and non-ideal models are independent from each other.

The ideal cycle is validated by evaluating COP/COP_{Carnot} and the cooling power as functions of ϕ to compare to literature figures from [16]. The cooling power is evaluated at four different ϕ_E . Similarly, the COP and the cooling power for the non-ideal cycle are evaluated at a fixed ϕ with varying I . Additionally, the non-ideal cycle is characterized by overpotentials evolution in the hot and cold electrochemical cells as functions of I , so these are also compared to results from [16] to validate the non-ideal model.

3. Results and Discussion

3.1 Model Validation and single-unit performance

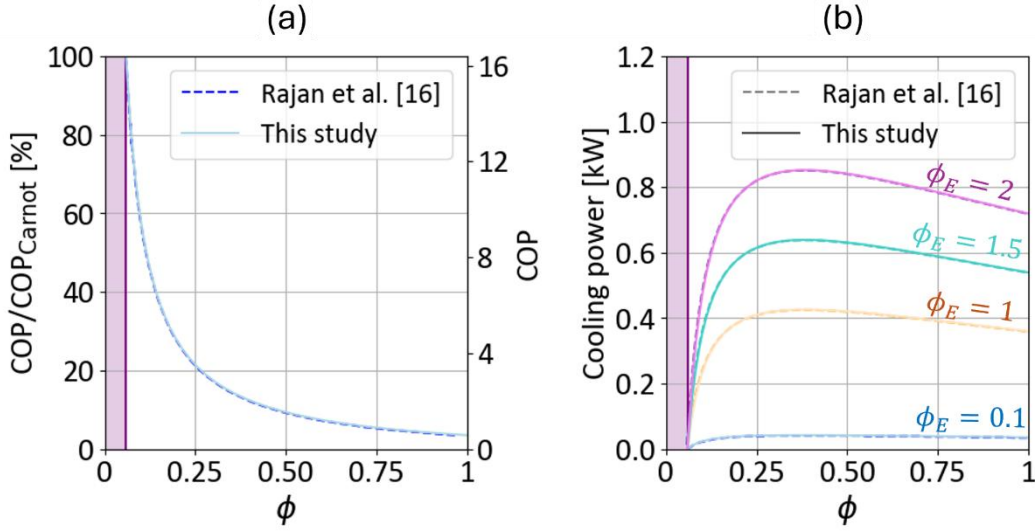


Figure 7 Ideal single-cycle COP_{ratio} (a) and cooling power (b) as functions of ϕ

The results used to validate the Python model for the ideal and non-ideal cycle are obtained for a unit operating with the input parameters in Table 1 and, as shown in Figure 7, in all cases the model results are in good accordance with the literature data [16], validating the code and allowing to use the ideal model for further analysis.

Assessing COP at different ϕ values (Figure 7a) highlights how the efficiency of the unit falls exponentially with increasing ϕ , when the entropic driving forces are high compared to the thermal inertia of the electrolyte. The behaviour is expected as a high ϕ implies large temperature difference between inlet and outlet of the electrochemical cells, resulting in T_2 and T_4 deviating from T_C and T_H (respectively). So, the processes in the heat exchangers deviate more from the isothermal heat exchanges of a Carnot cycle, creating the first layer of irreversibility according to the Clausius inequality, which formalises that the temperature gradient of heat exchange causes entropy generation (i.e. irreversibility). Therefore, in ideal and non-ideal cycles, maximum COP_{ratio} decreases with higher ΔT across heat exchangers.

It is important to notice that considerations in [16] are based on cooling operation as the refrigeration cycle is under study, but the conclusions equally apply to heating operations.

Analysing the cooling power in Figure 7b, it is evident that it increases sharply after cooling onset to a peak between $\phi = 0.25$ and $\phi = 0.5$, well outside of the “high-COP” regions, clearly introducing a trade-off between power and COP_{ratio} . The effects of high entropic forces in this case are crucial: high ϕ_E directly increase power. High ϕ_E can be achieved through operations at high currents, for a fixed reaction (i.e. fixed α).

Overall, a desirable operating region at high efficiencies implies a low power output, posing a barrier for practical usefulness in real heat pump systems and creating a need to decouple

COP and power. Even if ϕ in ideal cycles can be tuned by varying the volumetric flow rate of the electrolyte the maximum power region never coincides with the highest COP.

For the non-ideal model, the performance and the overpotentials (in both the hot and cold cells) are analysed against current to determine whether high I operations (and hence high power) can be the solution to operate at high-COP and high-power. The validating data include the performance (Figure 8a), the T-S diagrams for different current values (Figure 8b), and the overpotential development with I in the cold and hot cells (Figure 8c and d, respectively).

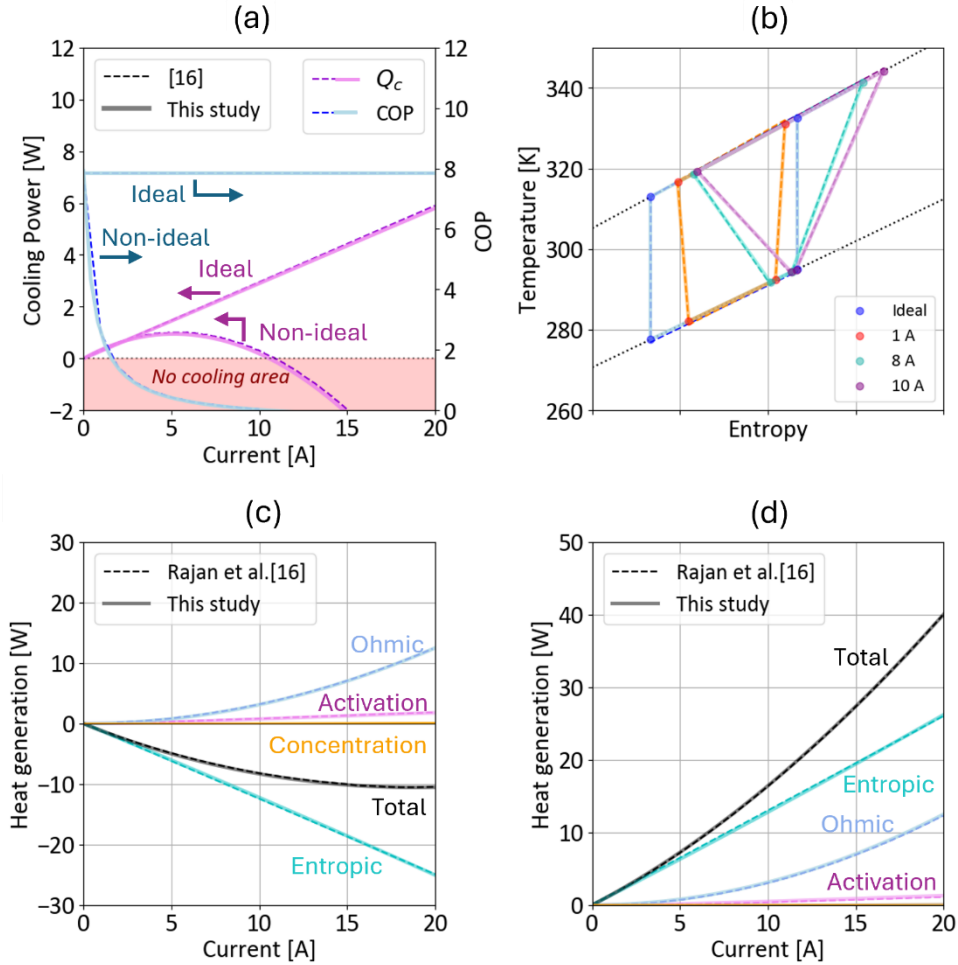


Figure 8 Non-ideal single-cycle performance (a), T-s diagram (b) and overpotentials evolution with current for cold (c) and hot (d) cells

Figure 8 shows how the model is overall validated. Noticeably, there is a minimal discrepancy for $I \gtrsim 8$ A in the non-ideal cooling power (Figure 8a). This is due to the result in the reference dataset [16] where $T_3 > T_c$ for $I \gtrsim 8$ A, implying a value for $\varepsilon_{cold\ HX} > 1$. This means that the fluid rejecting heat and entering the heat exchanger at T_c heated the electrolyte to a temperature higher than the fluid's own temperature, not attainable by processes in heat exchangers. However, this has minimal impact up to $I \approx 10$ A (hence on the region of interest). The present model corrects this behaviour and the Python model used in the project is determined validated and used for further conclusions.

The analysis is carried out using the iterative procedure described in 2.3.2 and the overpotentials for $I = [0, 20] A$ demonstrate a direct dependency on current (Figure 8c, d). In the cold cell, overpotentials effect become evident at $I \gtrsim 8 A$, where the total cooling power plateaus due to overpotential heat generation “neutralising” the entropic cooling (Figure 8c). At higher currents, this causes the temperature drops in the cold cell due to entropic cooling to be reduced to zero. Similarly in the hot cell, the overpotentials grow with current, generating additional heat and causing a large temperature rise in the hot cell. Recalling from above that a large electrolyte temperature at the inlet of the hot heat exchanger implies a large temperature range over which heat is transferred and so a large deviation from the Carnot cycle, which also is a source of irreversibility.

The COP and cooling power plot (Figure 8a) is further useful in confirming that, for a fixed ϕ , increasing I leads to an increase of ideal power without affecting ideal COP, but that this relationship falls once non-ideal conditions are introduced: the non-ideal COP falls exponentially with I while the power is strongly affected by the overpotential developments and increases almost ideally for low currents but then slows down and peaks at $I \approx 5 A$ (with this parameters), falling again as I increases and making operations at high I not viable. It is important to note that the analysis is for refrigeration but holds for heat pumps operations too.

The work presented in the literature and reproduced here makes evident that operation at low I and low ϕ are crucial for high efficiency. However, some factors not included in the literature are to be studied to thoroughly understand the behaviour of the real units in operation with respect to operating variables. One of these factors is the temperature range and lift of the system since all the analyses from [16] are at a fixed $T_H = 313 K$ and $T_C = 295 K$.

3.2 Temperature effects on single-cycle performance

Effect on COP

The effects of ΔT on the ideal COP_{ratio} of the single-cell unit are important to determine if an optimal temperature difference exists and, if so, which is it as well as understanding whether operating with the same ΔT at different T_H affects the COP_{ratio} . Hence, the two key variables assessed are the hot reservoir temperature T_H , and the temperature variation between reservoirs ΔT . Assessing T_H values of 295 K and 273 K accounts for cells operating close to the upper and lower bounds of domestic applications. ΔT is assessed in steps of 2 K with a maximum of value of 8 K to sweep the operating scenarios in which the cycles of the cascaded unit would operate. The results of the analysis are presented in Figure 9.

The similarity between the dashed and the solid lines indicated that the effect of T_H on the COP_{ratio} at a given ϕ and ΔT is minimal when operating in the regions of interest, and that the effect for lower ΔT is smaller. However, the ΔT has a significant effect on the ideal performance of the cycle: since the absolute ideal COP given by Equation 2.14 is weakly affected by T_r ,

(and hence ΔT), it results in an approximately constant value at a given ϕ . On the other hand, COP_{Carnot} is greatly affected by ΔT , falling with increasing temperature difference. The ratio COP/COP_{Carnot} then logically increases with increasing ΔT for a fixed ϕ .

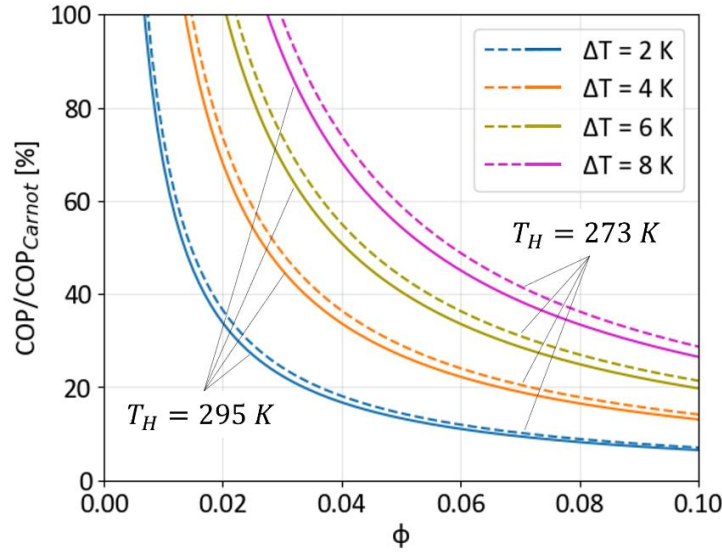


Figure 9 Effect of ΔT and T_H on ideal cycle COP

Even if the results just presented indicate that a high ΔT yields better COP_{ratio} , it is crucial to note how the *cooling onset* (i.e. the value of ϕ at which T_2 falls below T_3 and the heat exchange in the cold HX is in the right direction and the electrolyte effectively absorbs heat) moves at higher ϕ values with higher ΔT , which is equivalent to a higher ϕ requirement to achieve the same COP_{ratio} for higher ΔT . Recalling from the losses and performance analysis of a non-ideal single-cycle unit that lower ϕ is a desired operating point, a lower ΔT is desirable to “unlock” operations at lower ϕ . Additionally, smaller ϕ can be obtained by reducing I and thus reduce the losses in non-ideal operations.

Effect on losses

The overpotentials are the main cause of divergence from ideal operations for real cycles and the analysis of their development presented in Figure 8 (c and d) can be extended in a third dimension to factor in ΔT , using a fixed $T_H = 293\text{ K}$ representing a hypothetical building temperature of 20°C . Sweeping through temperature differences $0\text{ K} \leq \Delta T \leq 18\text{ K}$ to simulate air temperature changes, determines whether changes in T_C impact overpotentials.

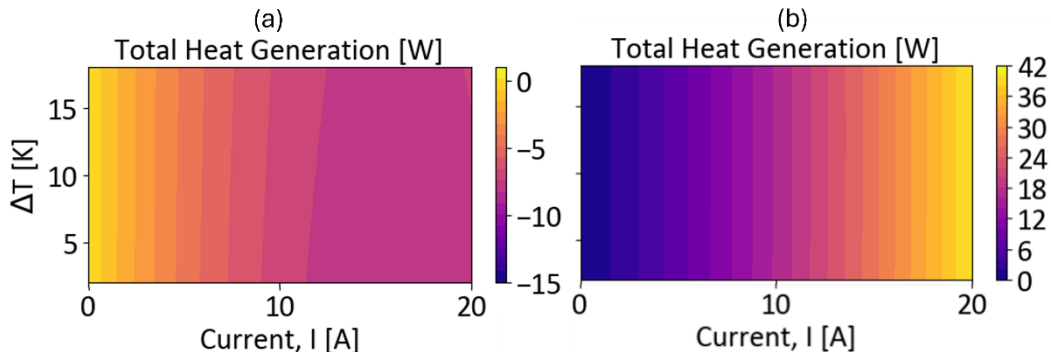


Figure 10 Total heat generation in cold (a) and hot (b) non-ideal cells

The results of the extended plots for total heat generation are presented in Figure 10 for both cells (cold in a and hot in b). The observed values tend to be relatively constant along the ΔT axis at each I , so the effect of temperature lift on the total heat generation is minor. Overall, the effect of ΔT on the overpotentials is considered negligible since the optimal operation region is at low currents ($I \lesssim 2 \text{ A}$), where no effect is relevant.

3.3 Single-cycle units performance limitations

The main result of the single cell analysis is a trade-off between COP and power, which has been investigated in the ideal case with respect to ϕ , and in the non-ideal case with respect to I . While the analysis indicated desired ranges of operations, a multi-objective optimisation algorithm is used to assess the effects of all the parameters simultaneously.

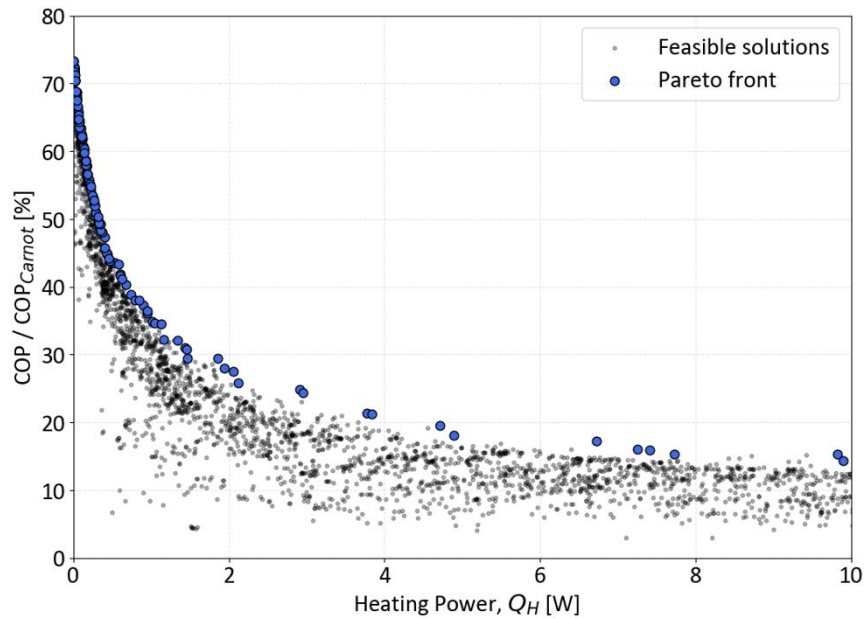


Figure 11 Performance trade-off for single-cycle units

The results of the investigation lead to the Pareto front between COP and heating power presented in Figure 11. The trade-off is evident and is a significant limiting factor for the development of the technology as the commercial units need to operate at high efficiency without being limited by power. While theoretically it would be possible to simply connect several single-cycle units to obtain a summative effect on power, this would lead to thousands of units even for a small domestic application, making the solution expensive and impractical. Therefore, it is crucial to (at least partially) reduce the trade-off by designing an “enhanced” unit able to deliver more power at acceptable COP values.

The Pareto front is useful to determine the desired operating ranges for the variables analysed and narrow down operations to high- COP_{ratio} regions. These regions are subsequently targeted to obtain high power without compromising COP. As logical, higher ε_{HX} and α yield a higher COP_{ratio} . However, instead of focusing on components or material

parameters, the analysis of ΔT , I , and ϕ is useful to assess operating conditions. As expected from Figure 9, higher ΔT results in higher COP_{ratio} (with the consequences and considerations discussed previously). Plotting only the Pareto points from Figure 11 with $COP_{ratio} > 45\%$ as functions of I and ϕ , the behaviour in Figure 12 is observed. Since Q_H is a linear function of I for a fixed ϕ (Equation 2.13), Figure 12a is just a scaled version of Figure 11, demonstrating how low I is better for COP_{ratio} . While Figure 12b is more scattered, it still shows how the very lower bound of the swept ϕ range is where the highest COP_{ratio} values occur. The behaviour in Figure 12 agrees with and generalises well the conclusions from Figure 7 and Figure 8.

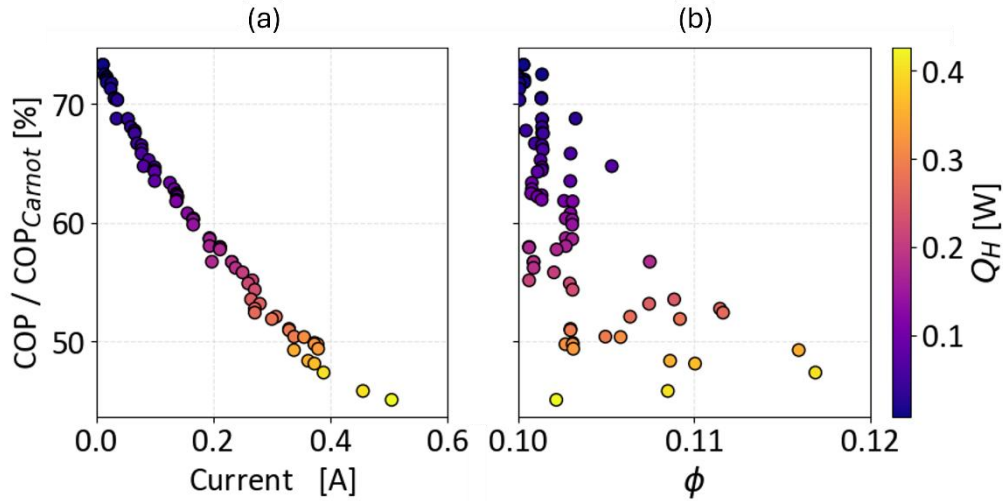


Figure 12 Performance from Pareto analysis as function of I (a) and ϕ (b)

The high COP_{ratio} regions emerging from the single-cycle analysis are used to determine the desired operating conditions. The three main objectives deriving from these are: (1) low ϕ values to operate as close as possible to the Carnot cycle (maximising ideal COP), (2) low I to minimise current-induced overpotential losses, and (3) low ΔT across cycles are preferred to reduce the minimum ϕ at which cooling onsets and operations become viable (even if COP_{ratio} increases with ΔT , in this case the ΔT is just used to “unlock” operations at lower ϕ , allowing to successfully operate at condition 1: low ϕ).

3.4 Enhanced performance with cascaded-cycle unit

3.4.1 Configuration selection tool

The cascaded solution could involve any number of cycles greater than 1, and the proposed unit is designed to be modular: the number of active cycles connected across the hot and cold reservoirs can be selected by the user between 1 and the total cycles comprising the unit (N). In this study, the goal is to demonstrate the working principle of the tool and analysing a unit with 5 available cycles enables coverage of the different behaviours of the unit depending on operating parameters and cycles configurations.

A representation of the cascaded unit with ideal cycles is in Figure 13, showing how each cycle operates at a different temperature with the ΔT across each cycle “building up” to reach ΔT_{max} . The key insight from Figure 8 that T_H has a small effect on the performance of a single cycle allows to simply treat the cascaded cycles as a set of identical cycles operating at different T_H (without the need to operate different cycles at different ΔT). The cycles in the figure are shown schematically on an arbitrary entropy axis just for illustrative purposes.

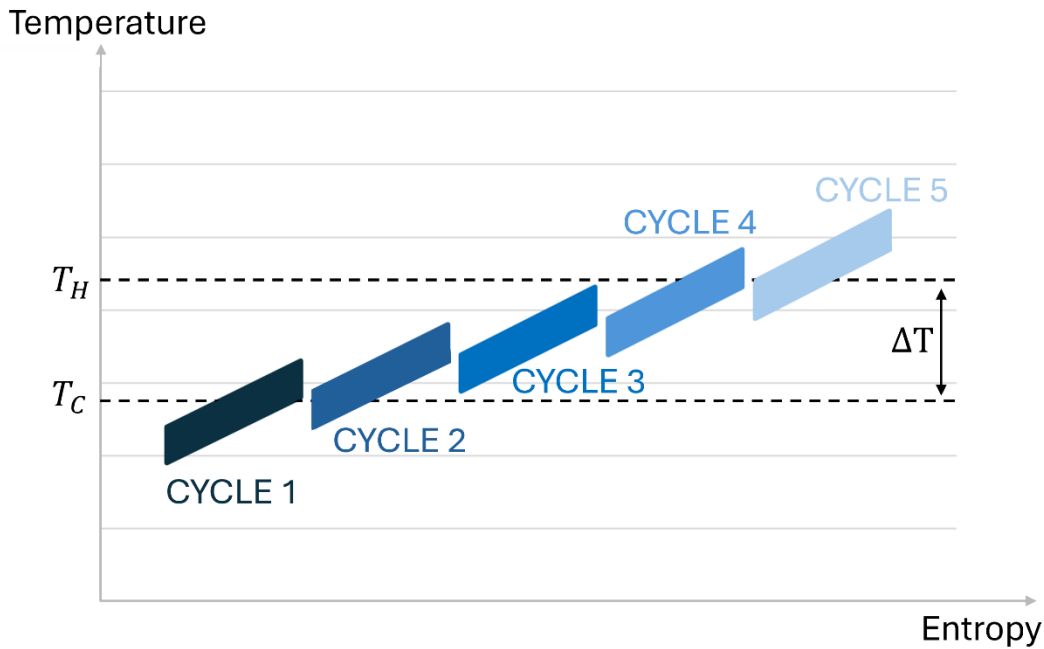


Figure 13 *T-S diagram of a cascaded unit with 5 cycles across ΔT*

Once the overall architecture is fixed, the modularity of the unit allows to select the operating configuration based on external conditions. The performance varies accordingly and the combination of COP and Q_H can be suited to specific situations such as start-up or constant power operations. To understand the general behaviour of the unit, the performance of the unit with 1, 2, 3, 4, and 5 cycles configurations as a function of ΔT with $T_H = 293\text{ K}$ (i.e. the ΔT reflects variations in T_C) is plotted in Figure 14.

While the trade-off between COP and power still exists, as can be seen by the inverse relative positions and trends of each configuration in the two plots of Figure 14, the cascaded-cycle unit enhances power and allows to operate with different configurations during different phases of the operation: at startup high power is needed and COP might be a secondary performance of interest so all the five cycles can be activated, while during continuous operations it is crucial to have high COP but low power is needed since the unit just needs to maintain the set building temperature and a two-cycle configuration could provide enough power at very high efficiency. In this simulation, operating four or five cycles is always viable (at every possible ΔT), while operations with less than four cycles are limited by the ΔT .

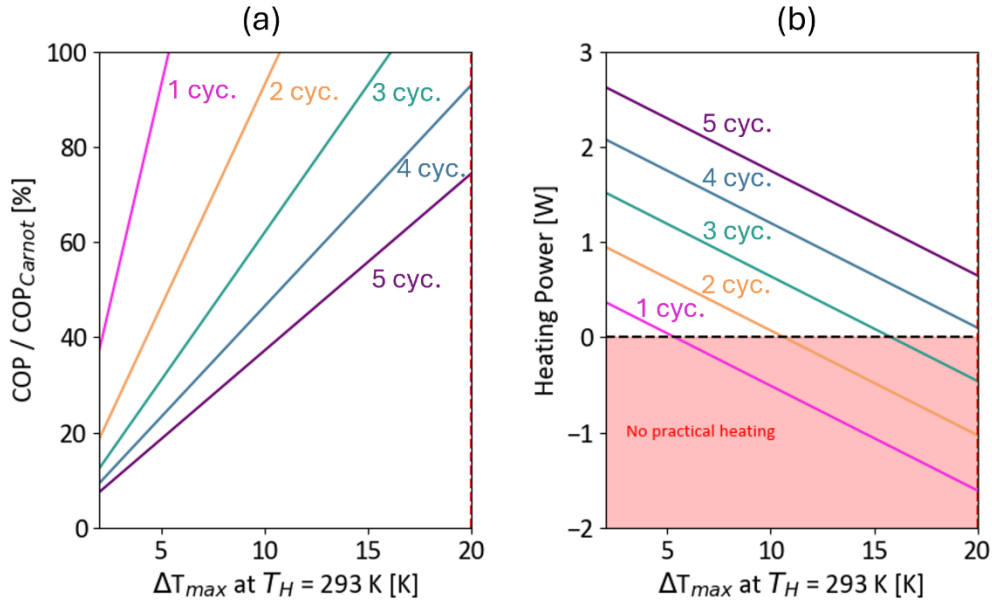


Figure 14 COP_{ratio} (a) and heating power (b) for 5 different configurations of a cascaded unit with maximum 5 active cycles

3.4.2 Cascaded-cycle unit case study

To better understand how the novel design compares to existing units proposed in the literature, it is sensible to provide a real-life example and assess differences between units.

The cascaded unit analysed comprises 5 ideal cascaded cycles. Each configuration of the cascaded cycle is evaluated for an external temperature T_C , given as input to the configuration tool. For this example, we investigate the performance at a temperature $T_C = 8^\circ\text{C} = 281$ K, which results in a $\Delta T_{max} = 12$ K and represents a realistic domestic scenario.

Table 2 summarises the operations at this ΔT_{max} for the possible number of active cycles in the cascaded unit presented in Section 3.4.3.

Table 2 Performance of the cascaded-cycles unit based on active cycles

Active cycles	COP / COP_{Carnot}	Heating power [W/unit] (heating occurs for >0)	Usable? (Yes/No)
2	—	−0.149	No
3	74%	0.421	Yes
4	56%	0.980	Yes
5	45%	1.529	Yes

While the next section compares this performance with the alternative single-cycle units, it is important to also consider the COP in absolute terms as, ultimately, one of the main factors that the customer cares about is the economic viability of the solution and absolute COP directly determines the electricity demand (and bills) from the heating power required. Since

the Carnot cycle has very high COP values at low ΔT , even with low COP_{ratio} values, the absolute COP could be much higher than that of a standard VCHP, leading to a great saving in electricity demand, hence reducing operating costs of the unit. The COPs for the three usable configurations are 18, 13, and 11 for the 3, 4, and 5 cycles (respectively). The ideal absolute COP values for the same context are around three to five times larger than the average real vapour compression heat pumps values (~ 3 to 4 [2]), and even if the real cascaded COP values are lower than the ideal values obtained, the margin is large enough to expect them to be greater than the real vapour compression heat pumps. This means that only a fraction of electrical power is needed to deliver the same amount of thermal power as the VCHP, resulting in a proportional cost saving for the user and endorsing the deployment of the technology for long term solutions. This result is also meaningful even in the context of electricity demand management by national grid operators: deploying cascaded electrochemical units reduces electricity demand to the grid, consequently reducing peak HP-related demand in a scenario where mass adoption of heat pumps is reality. Virtually, with a unit three times more efficient than the equivalent VCHP, three times the amount of heat pumps can be deployed with the same effect on peak electricity demand. This directly addresses some of the main concerns with widespread adoption of heat pumps and represents significant progress for the field. The significance of these results is clear once referring to the macro-objective of identifying a high-COP design with low-GWP working fluids.

However, the inverse relationship between COP and power still exists and therefore it is important to compare the performance to the equivalent single-cycle unit and quantify the enhancement to successfully answer the third and last research question.

3.4.3 Cascaded-cycle unit comparison with single-cycle options

To quantify the cascaded-cycle unit's benefits, it is crucial to define well what is considered to be the "equivalent" single-cycle unit. As introduced in Section 2.1, there are two possible configurations that can qualify as "equivalent" to a cascaded unit with N cycles (a visual example of which is presented in Figure 15 for a 3-cycle unit) based on what is set to be equal between the cascaded and single cycle units:

1. One single-cycle unit across the same ΔT_{max} with the same maximum and minimum unit temperatures as the cascaded unit: $T_{2 \text{ single-cycle}} = T_{2 \text{ coldest cascaded-cycle}}$ and $T_{4 \text{ single-cycle}} = T_{4 \text{ hottest cascaded-cycle}}$ ("Same T unit" in Figure 15),
2. N single cycles with the same I and ϕ as each of the cascaded cycles composing the cascaded unit (one cycle in "One out of N cycles with same I and ϕ " in Figure 15.)

And the performance of these two alternative options is compared to the cascaded-cycles unit with 3 active cycles in Table 3, operating in the same scenario as Section 3.4.2.

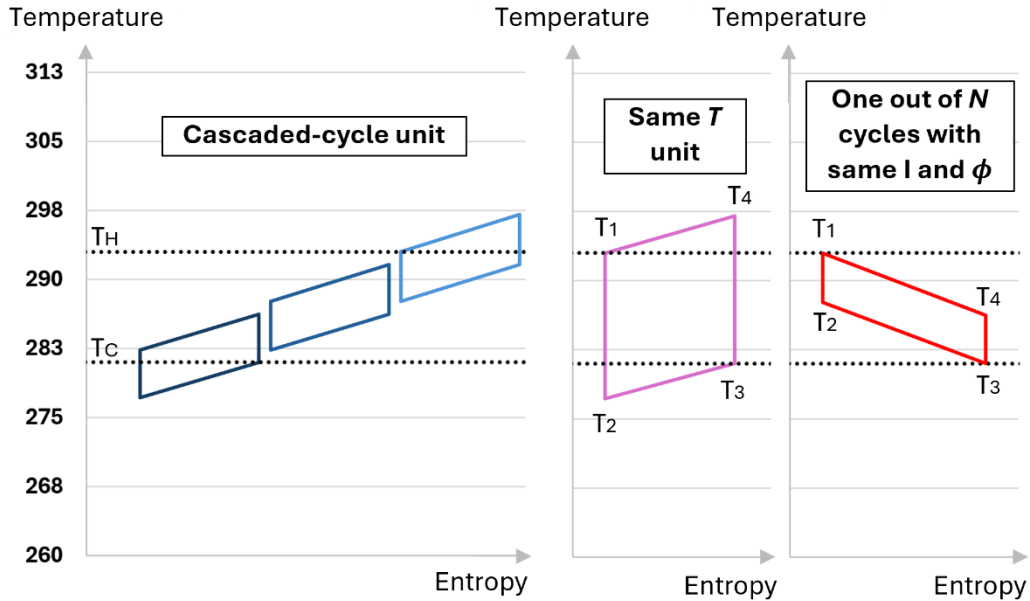


Figure 15 Cascaded unit compared to two single-cycle alternatives

Table 3 Performance comparison of the three alternative proposed

Option	COP/COP_{Carnot}	Heating power [W/unit]	Comments
Cascaded cycles unit	74%	0.42	Highest Q_H but COP is penalised
Same T unit	76%	0.15	Highest COP but very low Q_H
N cycles with same I and ϕ	No heating achievable	No heating achievable	ΔT in cells is too low due to ϕ value, $T_4 < T_1$ does not allow Q_H

It is clear that the cascaded cycles unit brings two major benefits compared to other possible solutions: (1) allows heating power to be much higher when the maximum and minimum unit temperatures are fixed (seen in the comparison between the “Same T unit” and the cascaded unit), and (2) “unlocks” heating operation at ϕ values that would not allow operations in a single-cell unit with the same current and ΔT (as seen by the comparison with the “same I and same ϕ ” unit).

Extending the analysis to 4 and 5 cascaded cycles, it is possible to compare with the equivalent single-cycle unit operating with the same maximum and minimum temperature and notice that the heating power increases by 190%, 278%, and 365% for 3, 4, and 5 cycles respectively. However, a small decrease in COP_{ratio} (around 2%) is also observed due to larger deviations from isothermal heat exchange in the heat exchangers.

From the previous conclusions that lower ϕ means higher COP_{ratio} values, and that low ϕ is obtained by “reduction” of I causing the losses of the non-ideal cycles to also be lower, it is clear that the cascaded design allows operation in the favourable ranges. In fact, it effectively achieves all the three operating requirements introduced by the assessment of single-cycle

units: ΔT_{max} is fractioned into compounded smaller ΔT , effectively allowing to operate each cycle at smaller ϕ and small I while still achieving high COP_{ratio} and higher heating power. This successfully addresses one of the largest barriers in the development and deployment of electrochemical units to phase-out VCHPs.

3.5 Limitations of the cascaded unit

The main limitation of the technology remains the relatively small power output related to operations at low currents, even if this issue has been strongly mitigated by the cascaded design. In fact, even if the trade-off between power and COP has been significantly reduced, operations at low currents imply by definition low power and it is an intrinsic feature of the electrochemical cycles that can never be fully avoided. Therefore, until material research allows to achieve higher values of α that can counteract the effects of low currents, the limitation can be seen as a deployment barrier of the technology. This implies that in a real device, it is crucial to take dimensions and cost into account and with the current state of materials development they could be prohibitive (however this analysis is out of the scope of this project).

The cascaded unit model presented employs ideal cycles and ideal heat exchangers between cycles. Even if the project is based on achieving low currents to obtain low losses under the assumptions that each of the cascaded cycles behaves like a single-cycle, it is crucial to quantify losses in real systems by addressing overpotential development in cascaded units in a similar fashion to what has been done in the single-cycle units and the compounded effect of such losses could grow in a prohibitive manner to operate in the desired regions, leading to a new barrier and limitation of the technology.

Finally, the operations example is only an approximation of real-world scenarios, and basing the analysis simply on the external temperature estimation is an extremely simplified approach. Many other factors affect the usage and performance of real heat pumps, and to design a real system they need to be accounted for. This example only uses the real-world approximation to locate the technology in a meaningful scenario and inform on the general performance in a realistic setting.

4. Conclusions and Future Work

This project fits within the research to identify and design a novel heat pump technology that can substitute vapour compression heat pumps to sustainably satisfy the predicted strong increase in demand for heat pumps over the following decades. The main required features of the novel technology are low-GWP working fluids and high cycle efficiency. Electrochemical heat pumps working on the Brayton cycle address these two points. However, the research in the field shows that performance of such systems is subject to a COP -power trade-off, posing a barrier to the widespread development of the units. The project aimed to address this limitation in electrochemical Brayton cycles for heat pumping by designing a novel configuration with enhanced performance. The solution presented in this work is an electrochemical heat pump unit based on cascaded Brayton electrochemical cycles. Cascading cycles partially offsets the trade-off and mitigates its effects, resulting in demonstrated power increases compared to equivalent single-cycle units of up to +365% with only 2% drops in COP_{ratio} .

The single-cycle units served as a starting point for the analysis, and the performance of ideal and non-ideal cycles has been investigated against operating conditions to identify desirable operational parameters. Low currents, low ϕ , and low temperature difference across cycles proved to be the best regions to operate at high COP_{ratio} , mainly due to higher ideal COP and low irreversibility from overpotentials, but resulted in low power outputs. Cascading cycles is used to increase the power of the unit while operating in the high COP regions. The cascaded unit reduces the temperature difference of each cycle and allows to operate each cycle with smaller ϕ and I while still producing power. Operations at low ϕ and I enhance the performance of the units increasing heating power and reducing efficiency losses.

Contextualising the main findings in a typical operation with an external-to-internal temperature difference of 12 K allows to visualise the enhanced performance of the cascaded cycle unit compared to the single cycle equivalent unit. The results with 3 cascaded cycles show heating power improvements of +190% with efficiency decreasing by 2%, extending the analysis to 4 and 5 cascaded cycles improves heating power by up to +365% while COP_{ratio} decreases by at most 2%. This shows a consistent enhancement of performance. The ideal absolute COP values for the same context are three to five times larger than the average real VCHP, a promising margin that benefits both the grid management and the consumer's wallet.

The implications of the findings are strong, as the new cascaded design directly provides an architecture with a higher combination of heating power and ideal COP_{ratio} for a given set of operating conditions. The unit can work at temperature ranges of interest for the domestic applications, making it a strong candidate as an alternative technology to vapour compression heat pumps. In addition, the new design effectively operates using aqueous electrolytes as

working fluids, addressing directly the barriers that limit the deployment of the technologies proposed in the literature due to sustainability concerns. However, the analysis is limited to the ideal case, with only qualitative considerations made on the operations in non-ideal contexts. While a higher ideal performance compared to the single-cycle unit combined with the operations at low currents is very promising for real systems performance, the effects of losses should be investigated further to truthfully position the technology in the landscape of alternatives to vapour compression heat pumps.

Key takeaways:

- The COP of ideal single-cycle units is determined by the temperature difference across heat exchange processes and constitutes the maximum achievable for the real systems, it is directly influenced by the non-dimensional parameter ϕ ;
- Non-ideal single-cycle units' performance is strongly affected by overpotential losses, due to operating currents, introducing a COP -heating power trade-off;
- The cascading-cycles unit allows operations at low ΔT across cycles, low ϕ , and low I , successfully addressing the barriers of single-cycle units and resulting in enhanced power up to +365% with minimal COP_{ratio} losses;
- The BEC HP design does not use any high-GWP refrigerant and demonstrated efficiencies 3 to 6 times those of the equivalent VCHP (in the case study analysed).

Future work should mainly be based on two areas: (1) material research to increase system performance through materials parameters rather than operating parameters, and (2) overpotential development and losses analysis in the cascaded cycle unit.

The second point is crucial and should be treated with high priority. A model which combines the architecture of the cascaded-cycle unit presented in this work with the losses evaluation algorithm used to describe non-ideal conditions in the single-cycle unit is the natural next step to locate the technology and the design in the landscape of alternative heat pumps to the standard vapour compression air source heat pump.

AI-Assisted writing declaration

The main AI contribution is in formatting of the plots and figures using Claude. I also used Claude for some minor rewriting of paragraphs and grammar check in long sections.

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6. Appendix

6.1 Cascaded Model and configuration tool code

```

#%% LIBRARIES
import numpy as np
import matplotlib.pyplot as plt
from matplotlib import cm
from matplotlib import colors
import matplotlib as mpl

fs = 11*1.5

mpl.rcParams.update({
    'font.family': 'sans-serif',
    'font.sans-serif': ['Calibri'],
    'font.size': fs,
    'axes.titlesize': fs,
    'axes.labelsize': fs,
    'xtick.labelsize': fs,
    'ytick.labelsize': fs,
    'legend.fontsize': fs,
})

#%% COP analysis for ideal cycle at different phi and DT
phi_pDT = np.arange(0.001,1+0.001,0.001)
DT_lines = [2, 4, 6, 8]
colors_lines = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728']

fig5, ax1 = plt.subplots(1, 1, figsize=(10*1.5/2.54, 8*1.5/2.54), constrained_layout=True)
ax1.set_title("COP / COP$_{Carnot}$", fontweight="bold")

for dt, col in zip(DT_lines, colors_lines):
    # TH = 293 K - solid lines
    TC_line = 295 - dt
    Tratio_line = 295 / TC_line
    Carnot_line = 1 / (Tratio_line - 1)
    lim_line = np.log(Tratio_line)

```

```

COP_line = (1 - (Tratio_line * np.exp(-phi_pDT))) / \
    (-Tratio_line - 1 + np.exp(phi_pDT) + Tratio_line * np.exp(-phi_pDT))
COPratio_line = COP_line / Carnot_line * 100
ax1.plot(phi_pDT, COPratio_line, color=col, lw=1.5, ls='-')

# TH = 273 K - dashed lines
TC_line2 = 273- dt
Tratio_line2 = 273/ TC_line2
Carnot_line2 = 1 / (Tratio_line2 - 1)
lim_line2 = np.log(Tratio_line2)
COP_line2 = (1 - (Tratio_line2 * np.exp(-phi_pDT))) / \
    (-Tratio_line2 - 1 + np.exp(phi_pDT) + Tratio_line2 * np.exp(-phi_pDT))
COPratio_line2 = COP_line2 / Carnot_line2 * 100
ax1.plot(phi_pDT, COPratio_line2, color=col, lw=1.5, ls='--', label=f" ΔT = {dt} K")

ax1.set_xlim([0, 0.1])
ax1.set_ylim([0, 100])
ax1.set_xticks(np.arange(0, 0.12, 0.02))
ax1.set_xlabel("φ")
ax1.set_ylabel("COP/COP$_{Carnot}$ [%]")
ax1.legend(loc="upper right", ncol=1)
ax1.grid(True, alpha=0.3)
plt.show(block=False)
fig5.savefig("COPIdealvsDTAndPhi.png")

# %% SECOND PART: DEFINING THE IDEAL MODELS
Cv = 4.18 *1e+6
TH = 293 #K room temperature ideal
TL = np.arange(20,205,5)/10
TC = TH-TL
Tratio = TC/TH
alp = 0.001
I = 1 #A
phi = 0.0185
V = (alp*I)/(phi*Cv)
COPcar = 1/(1-Tratio)

```

```
leng = len(TL)
```

```
Qh1c1 = np.zeros(leng)
```

```
Qh1c2 = np.zeros(leng)
```

```
Qh1c3 = np.zeros(leng)
```

```
Qh1c4 = np.zeros(leng)
```

```
Qh1c5 = np.zeros(leng)
```

```
Qc1c1 = np.zeros(leng)
```

```
Qc1c2 = np.zeros(leng)
```

```
Qc1c3 = np.zeros(leng)
```

```
Qc1c4 = np.zeros(leng)
```

```
Qc1c5 = np.zeros(leng)
```

```
COPcar1 = np.zeros(leng)
```

```
COPcar2 = np.zeros(leng)
```

```
COPcar3 = np.zeros(leng)
```

```
COPcar4 = np.zeros(leng)
```

```
COPcar5 = np.zeros(leng)
```

```
for i in range(leng):
```

```
    #1 CELL:
```

```
    CellsN = 1
```

```
    T11c1 = TH
```

```
    T21c1 = T11c1*np.exp(-phi)
```

```
    T31c1 = TC[i]
```

```
    T41c1 = T31c1*np.exp(phi)
```

```
    Tratio1c1 = T11c1/T31c1
```

```
    Qh1c1[i] = 2*I*alp*TC[i]/phi*(Tratio1c1-np.exp(phi))
```

```
    Qc1c1[i] = 2*I*alp*TC[i]/phi*(1-Tratio1c1*np.exp(-phi))
```

```
    W1c1 = Qh1c1[i]+Qc1c1[i]
```

```
    #2 CELLS:
```

```
    CellsN = 2
```

```
    T11c2 = TH
```

```
    T21c2 = T11c2*np.exp(-phi)
```

```
    T12c2 = T21c2
```

```
    T22c2 = T12c2*np.exp(-phi)
```

$$T32c2 = TC[i]$$

$$T42c2 = T32c2 * np.exp(phi)$$

$$T31c2 = T42c2$$

$$T41c2 = T31c2 * np.exp(phi)$$

$$Tratio1c2 = T11c2 / T31c2$$

$$Tratio2c2 = T12c2 / T32c2$$

$$Qh1c2[i] = 2 * I * alp * TC[i] / phi * (Tratio1c2 - np.exp(phi))$$

$$Qc1c2[i] = 2 * I * alp * TC[i] / phi * (1 - Tratio1c2 * np.exp(-phi))$$

$$Qh2c2 = 2 * I * alp * TC[i] / phi * (Tratio2c2 - np.exp(phi))$$

$$Qc2c2 = 2 * I * alp * TC[i] / phi * (1 - Tratio2c2 * np.exp(-phi))$$

$$W1c2 = Qh1c2[i] + Qc1c2[i]$$

$$W2c2 = Qh2c2 + Qc2c2$$

#3 CELLS:

$$CellsN = 3$$

$$T11c3 = TH$$

$$T21c3 = T11c3 * np.exp(-phi)$$

$$T12c3 = T21c3$$

$$T22c3 = T12c3 * np.exp(-phi)$$

$$T13c3 = T22c3$$

$$T23c3 = T13c3 * np.exp(-phi)$$

$$T33c3 = TC[i]$$

$$T43c3 = T33c3 * np.exp(phi)$$

$$T32c3 = T43c3$$

$$T42c3 = T32c3 * np.exp(phi)$$

$$T31c3 = T42c3$$

$$T41c3 = T31c3 * np.exp(phi)$$

$$Tratio1c3 = T11c3 / T31c3$$

$$Tratio2c3 = T12c3 / T32c3$$

$$Tratio3c3 = T13c3 / T33c3$$

$$Qh1c3[i] = 2 * I * alp * TC[i] / phi * (Tratio1c3 - np.exp(phi))$$

$$Qc1c3[i] = 2 * I * \alpha_p * TC[i] / \phi * (1 - Tratio1c3 * np.exp(-\phi))$$

$$Qh2c3 = 2 * I * \alpha_p * TC[i] / \phi * (Tratio2c3 - np.exp(\phi))$$

$$Qc2c3 = 2 * I * \alpha_p * TC[i] / \phi * (1 - Tratio2c3 * np.exp(-\phi))$$

$$Qh3c3 = 2 * I * \alpha_p * TC[i] / \phi * (Tratio3c3 - np.exp(\phi))$$

$$Qc3c3 = 2 * I * \alpha_p * TC[i] / \phi * (1 - Tratio3c3 * np.exp(-\phi))$$

$$W1c3 = Qh1c3[i] + Qc1c3[i]$$

$$W2c3 = Qh2c3 + Qc2c3$$

$$W3c3 = Qh3c3 + Qc3c3$$

#4 CELLS:

$$CellsN = 4$$

$$T11c4 = TH$$

$$T21c4 = T11c4 * np.exp(-\phi)$$

$$T12c4 = T21c4$$

$$T22c4 = T12c4 * np.exp(-\phi)$$

$$T13c4 = T22c4$$

$$T23c4 = T13c4 * np.exp(-\phi)$$

$$T14c4 = T23c4$$

$$T24c4 = T14c4 * np.exp(-\phi)$$

$$T34c4 = TC[i]$$

$$T44c4 = T34c4 * np.exp(\phi)$$

$$T33c4 = T44c4$$

$$T43c4 = T33c4 * np.exp(\phi)$$

$$T32c4 = T43c4$$

$$T42c4 = T32c4 * np.exp(\phi)$$

$$T31c4 = T42c4$$

$$T41c4 = T31c4 * np.exp(\phi)$$

$$Tratio1c4 = T11c4 / T31c4$$

$$Tratio2c4 = T12c4 / T32c4$$

$$Tratio3c4 = T13c4 / T33c4$$

$$Tratio4c4 = T14c4 / T34c4$$

$$Qh1c4[i] = 2 * I * \alpha_p * TC[i] / \phi * (Tratio1c4 - np.exp(\phi))$$

$$Qc1c4[i] = 2 * I * \alpha_p * TC[i] / \phi * (1 - Tratio1c4 * np.exp(-\phi))$$

$$Qh2c4 = 2 * I * \alpha_p * TC[i] / \phi * (Tratio2c4 - np.exp(\phi))$$

$$\begin{aligned} Qc2c4 &= 2 * I * alp * TC[i] / phi * (1 - Tratio2c4 * np.exp(-phi)) \\ Qh3c4 &= 2 * I * alp * TC[i] / phi * (Tratio3c4 * np.exp(phi)) \\ Qc3c4 &= 2 * I * alp * TC[i] / phi * (1 - Tratio3c4 * np.exp(-phi)) \\ Qh4c4 &= 2 * I * alp * TC[i] / phi * (Tratio4c4 * np.exp(phi)) \\ Qc4c4 &= 2 * I * alp * TC[i] / phi * (1 - Tratio4c4 * np.exp(-phi)) \\ W1c4 &= Qh1c4[i] + Qc1c4[i] \\ W2c4 &= Qh2c4 + Qc2c4 \\ W3c4 &= Qh3c4 + Qc3c4 \\ W4c4 &= Qh4c4 + Qc4c4 \end{aligned}$$

#5 CELLS:

CellsN = 5

T11c5 = TH

T21c5 = T11c5 * np.exp(-phi)

T12c5 = T21c5

T22c5 = T12c5 * np.exp(-phi)

T13c5 = T22c5

T23c5 = T13c5 * np.exp(-phi)

T14c5 = T23c5

T24c5 = T14c5 * np.exp(-phi)

T15c5 = T24c5

T25c5 = T15c5 * np.exp(-phi)

T35c5 = TC[i]

T45c5 = T35c5 * np.exp(phi)

T34c5 = T45c5

T44c5 = T34c5 * np.exp(phi)

T33c5 = T44c5

T43c5 = T33c5 * np.exp(phi)

T32c5 = T43c5

T42c5 = T32c5 * np.exp(phi)

T31c5 = T42c5

T41c5 = T31c5 * np.exp(phi)

Tratio1c5 = T11c5 / T31c5

Tratio2c5 = T12c5 / T32c5

```

Tratio3c5 = T13c5/T33c5
Tratio4c5 = T14c5/T34c5
Tratio5c5 = T15c5/T35c5
Qh1c5[i] = 2*I*alp*TC[i]/phi*(Tratio1c5-np.exp(phi))
Qc1c5[i] = 2*I*alp*TC[i]/phi*(1-Tratio1c5*np.exp(-phi))
Qh2c5 = 2*I*alp*TC[i]/phi*(Tratio2c5-np.exp(phi))
Qc2c5 = 2*I*alp*TC[i]/phi*(1-Tratio2c5*np.exp(-phi))
Qh3c5 = 2*I*alp*TC[i]/phi*(Tratio3c5-np.exp(phi))
Qc3c5 = 2*I*alp*TC[i]/phi*(1-Tratio3c5*np.exp(-phi))
Qh4c5 = 2*I*alp*TC[i]/phi*(Tratio4c5-np.exp(phi))
Qc4c5 = 2*I*alp*TC[i]/phi*(1-Tratio4c5*np.exp(-phi))
Qh5c5 = 2*I*alp*TC[i]/phi*(Tratio5c5-np.exp(phi))
Qc5c5 = 2*I*alp*TC[i]/phi*(1-Tratio5c5*np.exp(-phi))
W1c5 = Qh1c5[i]+Qc1c5[i]
W2c5 = Qh2c5+Qc2c5
W3c5 = Qh3c5+Qc3c5
W4c5 = Qh4c5+Qc4c5
W5c5 = Qh5c5+Qc5c5

Wtot1 = W1c1
Wtot2 = W1c2+W2c2
Wtot3 = W1c3+W2c3+W3c3
Wtot4 = W1c4+W2c4+W3c4+W4c4
Wtot5 = W1c5+W2c5+W3c5+W4c5+W5c5
COPh1 = Qh1c1[i]/Wtot1
COPh2 = Qh1c2[i]/Wtot2
COPh3 = Qh1c3[i]/Wtot3
COPh4 = Qh1c4[i]/Wtot4
COPh5 = Qh1c5[i]/Wtot5
COPcar1[i] = COPh1/COPcar[i]
COPcar2[i] = COPh2/COPcar[i]
COPcar3[i] = COPh3/COPcar[i]
COPcar4[i] = COPh4/COPcar[i]
COPcar5[i] = COPh5/COPcar[i]

```

```
###
```

```
while True:
```

```

try:
    TL_op = float(input(f"Enter the operating Temperature Lift TL (between {TL.min():.1f}
and {TL.max():.1f} K): "))
    if TL.min() <= TL_op <= TL.max():
        break
    else:
        print(f" ⚠ Please enter a value between {TL.min():.1f} and {TL.max():.1f}.")
except ValueError:
    print(" ⚠ Invalid input. Please enter a number.")

```

```

colors = ['#e63946', '#f4a261', '#2a9d8f', '#457b9d', '#6a0572']
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(21/2.54, 12/2.54), constrained_layout=True,
gridspec_kw={'wspace': 0.1})

```

```

ax1.plot(TL, COPcar1*100, label='1 cell', color=colors[0])
ax1.plot(TL, COPcar2*100, label='2 cells', color=colors[1])
ax1.plot(TL, COPcar3*100, label='3 cells', color=colors[2])
ax1.plot(TL, COPcar4*100, label='4 cells', color=colors[3])
ax1.plot(TL, COPcar5*100, label='5 cells', color=colors[4])
ax1.set_ylim(0, 100)
ax1.set_xlim(2, 20)
ax1.set_xlabel(r'$\Delta T_{\max}$ at $T_H = 293$ K [K]')
ax1.set_ylabel('COP / COP$_{\text{Carnot}}$ [%]')
ax1.axvline(x=TL_op, color='red', linestyle='--', linewidth=1.5, label=f'TL = {TL_op:.1f} K')

```

```

lines = [(-Qh1c1, '1 cell'), (-Qh1c2, '2 cells'), (-Qh1c3, '3 cells'),
         (-Qh1c4, '4 cells'), (-Qh1c5, '5 cells')]
ax2.axhspan(-2, 0, color='red', alpha=0.25, zorder=0)
ax2.text(3, -1, 'No practical heating',
        ha='left', va='center', color='Red', fontsize=11)
for (y, lbl), c in zip(lines, colors):
    ax2.plot(TL, y, label=lbl, color=c)
ax2.plot([2, 20], [0, 0], 'k--')
ax2.set_xlim(2, 20)
ax2.set_ylim(-2, 3)

```

```

ax2.set_xlabel(r'$\Delta T_{\max}$ at $T_H = 293$ K [K]')
ax2.set_ylabel('Heating Power [W]')
ax2.axvline(x=TL_op, color='red', linestyle='--', linewidth=1.5, label=f'TL = {TL_op:.1f} K')

plt.show()

# %%
# Summary Table
# Convert TL_op to array index: TL starts at 2K in steps of 0.5K
idx = int(round((TL_op - 2.0) / 0.5))

Qh_values = [-Qh1c1[idx], -Qh1c2[idx], -Qh1c3[idx], -Qh1c4[idx], -Qh1c5[idx]]
COP_values = [COPcar1[idx], COPcar2[idx], COPcar3[idx], COPcar4[idx], COPcar5[idx]]
COPact_values = [COPh1, COPh2, COPh3, COPh4, COPh5]

print(f"\n{'-'*85}")
print(f" Operating Summary at TL = {TL_op:.1f} K")
print(f"{'-'*85}")
print(f"  {'Cycles':<8} {'Heating Power per unit [W/unit]':>25} {'COP/COP_Carnot':>15}
{'COP':>9} {'Usable?':>14}")
print(f"{'-'*85}")

viable_found = False
for n, (Qh, cop, copa) in enumerate(zip(Qh_values, COP_values, COPact_values),
start=1):
    usable = 0.40 <= cop <= 1.00
    if usable:
        viable_found = True
        cop_str = f"{cop*100:>14.1f}%"
    else:
        cop_str = f"{'-':>15}"
    flag = "✓ YES" if usable else "X NO"
    print(f" {n} cycle{'s' if n>1 else ''}<19> {Qh:>14.3f} {cop_str} {copa:10.1f} {flag:>11}")

print(f"{'-'*85}")

```

```
if not viable_found:
    print(" ⚠ No viable configuration at this temperature lift.")
else:
    best_idx = max(
        (i for i, cop in enumerate(COP_values) if 0.40 <= cop <= 1.00),
        key=lambda i: Qh_values[i]
    )
    print(f" Highest power option: {best_idx+1} cell{'s' if best_idx>0 else ''} "
          f"→ {Qh_values[best_idx]:.3f} W/unit | {COP_values[best_idx]*100:.1f}% of
Carnot")

print(f"{' '*85}\n")
```